BlenderMesh2STEP

Turn Blender meshes into real CAD STEP solids

User Guide — version 0.11.0

2026-07-11

Abstract

BlenderMesh2STEP is a Blender 4.2+ extension that converts triangle meshes into genuine STEP AP242 B-rep solids — the kind FreeCAD, Fusion 360, SolidWorks and Onshape open as editable, measurable, manufacturable geometry. You select the faces of a feature, the extension fits the exact analytic surface (plane, box, cylinder, cone, sphere, torus, fillet) by least squares — typically to machine precision on clean Blender-authored meshes — and the exporter writes a true solid, with real holes and pockets cut by boolean operations. This guide covers installation, the reverse (mesh → CAD) workflow, forward parametric modeling, exporting, and verifying the result.

Contents

1 Introduction

1.1 The problem

A mesh has thrown away the *intent* you want back: a 20 mm cylinder and a 64-sided prism are identical triangle soups. Exporting STL (or tessellated STEP) hands a machine shop a faceted blob they cannot measure, fillet, or edit.

BlenderMesh2STEP does not guess — **you point, it reconstructs**. You tell it "these faces are a cylinder," and it recovers the exact analytic surface by least-squares fitting. The result is a true boundary-representation (B-rep) solid with real `CYLINDRICAL_SURFACE` entities, not an approximation. On clean meshes the fits land at $\sim 10^{-8}$ RMS — effectively exact.

1.2 Two workflows

Reverse (mesh → CAD)
Select the faces of one feature in Edit Mode, fit a primitive, repeat per feature, export. This is the main workflow and the subject of Section 4.

Forward (build → CAD)
Add parametric STEP-primitive solids directly from the **Build — STEP Primitives** panel. These are STEP-exact *by construction*; the viewport mesh is only a preview. See Section 6.

Both workflows feed the same *feature stack* and the same exporter, and can be mixed freely — e.g. fit a base plate from a mesh, then add a forward-built cylinder as a *Subtract* feature to drill a hole.

1.3 Two export backends

OCCT kernel (recommended)
Uses OpenCASCADE — the kernel FreeCAD is built on — installed on demand from the panel (Section 2.3). This is the path that can *cut holes* (booleans), merge bodies into one watertight solid, heal gaps, and validate the result (per-solid volume and topology checks).

Pure-Python writer (fallback)
Zero dependencies. Writes real analytic AP242 solids, but has **no boolean kernel**: subtractive features cannot be cut from the part (Section 7.5).

> **Important**
>
> If your part has holes, pockets, counterbores, or must be a single watertight body, install the OCCT kernel. Without it a "hole" can only be marked, skipped, or written as filled material — never actually cut.

2 Installation

2.1 Requirements

- Blender **4.2 or newer** (the extension uses the 4.2+ extension format).
- No other dependencies for fitting and the pure-Python STEP writer.
- Optionally, the OCCT kernel (∼100 MB, one click) for booleans, watertight sewing, and validation.

2.2 Installing the extension

1. Download `reverse_mesh-0.11.0.zip` from the project's GitHub Releases page.
2. In Blender: **Edit** → **Preferences** → **Get Extensions** → (top-right dropdown) **Install from Disk...** and pick the zip.
3. Enable it if it is not enabled automatically. The **Reverse** tab appears in the 3D-viewport sidebar (press `N`).

2.3 Installing the OCCT kernel

In the **Reverse** tab, open **Fitted Features** → **Export** and click **Install OCCT**. This downloads the `cadquery-ocp` binding into the extension's own directory (it does not touch your system Python). When it is ready the panel shows **OCCT ready**. All boolean and watertightness options in the export dialog are marked *(OCCT only)* and silently do nothing on the pure-Python path.

3 Quick start: the 60-second workflow

1. Select your mesh object, enter **Edit Mode**, open the **Reverse** tab in the sidebar (`N`).
2. Choose a **Primitive** (or leave **Auto-detect**) and a **Role**: *Add* for material, *Subtract* for a hole/pocket cutter.
3. Select the faces of *one* feature. Click one face of a wall and use **Select Similar** to grow the selection across the whole surface (it stops at sharp edges).
4. Click **Fit Primitive to Selection**. The fit lands in the **Fitted Features** stack with its RMS error; the optional heatmap colours the faces by deviation. Repeat for each feature.
5. Click **Export STEP (AP242)**, review the options (Section 7), export, and read the **Validation Report** panel to confirm volumes and watertightness.

4 The reverse workflow in detail

4.1 Primitives you can fit

Primitive	Recovers	Notes
Plane	flat face with extent	exported as a bounded planar patch
Box	oriented cuboid (rotation too)	fit from several flat faces at once
Cylinder	axis, radius, height	the workhorse for holes and bosses
Cone	apex, half-angle, radii	full cones and frustums
Sphere	centre, radius	
Torus	axis, major + minor radius	best on full rings
Fillet	edge round → partial cylinder	radius + arc extent; arcs $< 330°$
Extrude	prism: planar line/arc profile + height	select the whole prism incl. caps
Auto	best of the above	normal-agreement gate + simplicity tie-break

Table 1: Fittable primitives. All fits report RMS and maximum error.

The **Extrude** fit recovers the archetypal Blender-authored mechanical part: a planar outline of lines and arcs swept along an axis (brackets, plates, slots, bosses). Select the *entire* prism — side walls and both end caps — and the fitter recovers the axis, the height, and an exact line/arc profile (tangent arcs such as slot ends are recognised as true arcs, not chains of facets). It exports as a genuine B-rep solid on both backends and can act as a *Subtract* cutter (a milled pocket).

Auto-detect tries every primitive and keeps the best fit, preferring the *simplest* primitive when several explain the selection equally well (a flat patch becomes a plane, not a giant sphere). The runner-up candidates are shown for the active feature so you can see how close the call was.

4.2 Selecting a feature's faces

- **Select Similar** grows the selection from the active face across edges whose face-normal angle is below the *crease angle* — one click on a cylindrical wall selects the whole wall and stops at the caps.
- **Segment regions** (in the main panel) splits a multi-surface selection into smooth-connected regions and fits each separately — e.g. a whole cube selection becomes 6 planes; a full cylinder becomes side + 2 caps.

4.3 Fit quality controls

Fit-quality heatmap
: Colours the selected faces green → red by deviation from the fitted surface, so a stray chamfer or mis-picked face is obvious instead of buried in an average.

Reject outliers (RANSAC)
: Trims faces that deviate from the consensus surface and refits on the rest. Use when a selection is slightly dirty (an odd triangle, the start of a chamfer).

Snap dimensions
: Rounds fitted radii/lengths to a nice value (e.g. 19.98 → 20.0) when they are already very close, so the STEP carries manufacturable numbers. Presets 0.1 / 0.5 / 1.0 or a custom step.

Tolerance
: The RMS threshold (as a fraction of region size) above which the fit is flagged as poor.

4.4 Propagate Pattern: fit one hole, get the rest

Fit *one* hole (a subtractive cylinder), then click **Propagate Pattern**. The extension scans the mesh for every matching hole — same radius and axis direction within tolerance — and fits them all with the seed's settings. Bolt circles, linear arrays and mirrored holes are recognised and labelled (circular / linear / scattered).

4.5 Whole-mesh auto-decompose (experimental)

Two heavier, whole-mesh paths exist for when you want a first pass without hand-selecting anything. Both are marked red in the UI because they optimise over the *entire* mesh and can take a while:

Auto-Decompose Whole Mesh
: Surface decomposition: sweeps coarse→fine crease angles, builds competing primitive hypotheses, and selects a set by global energy minimisation (data fit + coverage + a per-primitive complexity cost + boundary smoothness). Faces no primitive explains can be kept as a `Reverse_Leftover` faceted patch so the exported STEP still contains the complete part.

Auto-Decompose Solid (Boolean)
: Volumetric decomposition: samples the interior volume and greedily covers it with inscribed spheres, cylinders and boxes, producing an additive union of solids. Export with the OCCT **Merge into one solid** option.

Treat both as a convenience to correct, not a promise: review the resulting feature set, delete bad members, and re-fit where needed.

5 The feature stack

Every fit and every forward-built primitive lands in **Fitted Features**. Each entry shows its boolean role (`+/-`), kind, a summary with dimensions, and its RMS (or "exact" for built primitives). Prefixes mark machine-generated sets: `[A]` surface auto-decompose, `[S]` solid decompose, `[B]` forward-built.

From the stack you can, non-destructively:

- **Reorder** features (arrows). Order matters with *Booleans in stack order* at export: an Add placed after a cut refills it (e.g. a boss inside a pocket).
- **Re-fit** a feature from its remembered source faces after editing the mesh.
- **Toggle Add / Subtract**, and for subtractive features choose **Through** (cutter overshoots both ends so the hole opens cleanly) or **Blind** (keeps the pocket depth exact; the open end is auto-detected).
- **Tag threads**: on a cylinder/cone, enter e.g. `M8x1.25`. The designation is written into the STEP product name and the PMI sidecar.
- **Hole presets**: on a subtractive cylinder choose **Counterbore** (recess radius + depth) or **Countersink** (recess radius + angle); the recess is cut as an extra coaxial cutter on the OCCT path.
- **Select** the feature's generated object in the viewport; **delete** single features or **clear** whole sets.

The stack persists in the `.blend` file; the authoritative parameters of each feature live on its generated object, so features survive reloads and renames of the session list.

6 Forward modeling: Build — STEP Primitives

The **Build** panel adds solids that are STEP-exportable by construction: **Box, Cylinder, Cone (frustum), Sphere, Torus, Extrude (N-gon prism)**. The exact analytic parameters are stored on the object; the viewport mesh is just a preview tessellation. For the extruded prism, pick the side count when adding; the circumradius and height stay live-editable and the profile follows.

- Pick a primitive and a **Role** (Add or Subtract — a forward-built cylinder makes a perfect drill), then **Add Primitive**.
- With the object active, its dimensions appear as live-editable fields; editing one regenerates the mesh immediately.
- If you move/rotate the object, that transform is carried into the export. If you *scale* it in Object Mode, the panel offers **Bake Scale into Parameters** — click it so the stored dimensions match what you see. Non-uniform scale on curved primitives is flagged: it is not representable as an analytic STEP surface.
- If the mesh drifts out of sync with the stored parameters (e.g. you edited vertices by hand), the panel warns and offers **Rebuild from Parameters**.

7 Exporting STEP

Click **Export STEP (AP242)** in the **Fitted Features** panel. The file dialog carries the export options:

Option	Meaning
Backend	**Auto** (OCCT if installed, else pure Python), or force either.
Merge into one solid	Fuse all Add solids into a single body (OCCT).
Booleans in stack order	Apply Add/Subtract in feature-stack order so a later Add refills a cut. Off = fuse all Adds first, then cut (OCCT).
Cutter overshoot	Extend cutters by this fraction at each end so through holes open cleanly on coplanar faces (OCCT).
Auto-stitch shared edges	Unify coincident faces so abutting features share real edges (OCCT).
Make watertight + Sew tolerance	Sew all faces, build a closed solid, heal gaps up to the tolerance; reports remaining open edges (OCCT).
Cutters (pure Python)	What to do with Subtract features when there is no kernel: **Include marked** (red `cutter:` reference solids — default), **Skip**, or **Include as solids** (holes appear filled).

Table 2: Backend/boolean export options.

7.1 Backend and boolean options

7.2 Units and scale

The **Unit** option sets the length unit declared inside the STEP file (mm / m / inch); it defaults to your scene's display unit. **Scale from Scene units** derives the coordinate scale from Blender's unit settings — e.g. a metric scene at Unit Scale 0.001 with STEP unit mm gives `1 BU` → `1 mm`. The dialog always shows the *effective scale* live; check it before exporting. Choose **Manual** to force a factor. Scenes with unit system *None* pass coordinates through unchanged.

7.3 Scope and extras

- **Selected only** exports just the selected Reverse objects.
- **Feature set** filters by group: All / Manual / Built / Auto / Solid — useful when an auto-decompose set coexists with hand fits.
- **Include leftover patches** writes the unexplained-face patches so the STEP contains the complete part (as faceted planar faces).
- **Write PMI sidecar** adds `.pmi.json` and `.pmi.csv` next to the STEP with every feature's dimensions (radii, diameters, lengths, angles, threads) and pairwise relationships (axis angles, hole spacing) — handy for inspection sheets.
- **Embed semantic PMI** embeds AP242 `DIMENSIONAL_SIZE` entities so CAD reads diameters/lengths as queryable PMI (pure-Python writer only).

7.4 The validation report

After an OCCT export, the **Validation Report** sub-panel lists every solid with its volume and validity, plus a watertightness line (open-edge count if not). Read it every time: *valid + watertight + plausible volume* is your evidence the file will survive a CAD import and a machine shop's quote. The pure-Python path cannot validate; it says so instead of pretending.

7.5 Limits of the pure-Python writer

The built-in writer emits real analytic solids (box, cylinder, cone, sphere, torus), bounded plane patches, trimmed partial-cylinder fillet patches, and faceted leftover patches — with per-feature colours. It cannot: perform booleans (no holes), merge bodies, sew to watertight, or validate. Overlapping Add bodies are written as separate interpenetrating solids in one part.

8 Verification and troubleshooting

8.1 Verifying a STEP file

- Open it in FreeCAD: it should appear as solid bodies, not a mesh. **Part** workbench → **Check geometry** should report no errors for watertight exports.
- Measure a known dimension (a hole diameter, a plate thickness) — the point of this tool is that those numbers are exact.
- Compare the reported volume in the **Validation Report** against an expectation ($\pi r^2 h$ for a cylinder, etc.).

8.2 Common issues

Symptom	Likely cause / fix
Part imports 1000× too big/small	Unit mismatch. Check the *effective scale* line in the export dialog (Section 7.2).
Holes are filled in the imported part	Exported on the pure-Python path with **Include as solids**, or OCCT not installed. Install OCCT; keep **Cutters** on **Include marked**.
Hole doesn't break through the surface	Increase **Cutter overshoot**, or set the feature's cut mode to **Through**.
Watertight pass reports open edges	Leftover/fillet patches only meet fitted surfaces within the fit deviation. Raise **Sew tolerance** slightly, or exclude leftovers.
Fit has a high RMS	The selection includes faces from another surface. Use the heatmap to spot them, or enable **Reject outliers**.
Auto-detect picks the wrong primitive	Select more of the surface (partial patches are ambiguous), or choose the primitive explicitly — the runner-up list shows how close the call was.
Torus/fillet fit is approximate	Known limitation on partial patches; select as much of the ring/round as possible. Fillet arcs must be $< 330°$.
STEP file is huge	Leftover patches carry many triangles. Remesh, or disable **Include leftover patches**.

Table 3: Troubleshooting quick reference.

Tip

Model with intent in Blender and the fits will be exact: keep flat things flat, use enough segments on curved surfaces, apply scale (`Ctrl+A`) before fitting, and prefer one feature per surface.

9 Roadmap

Development continues along the plan in `PLAN.md` at the repository root: an OCCT-first experience, extruded-profile and revolve recovery (genuine sketch-based solids), fillets and chamfers applied as real blends, multi-body STEP with colours, and — longer term — links from the feature stack into Blender's simulation systems and an optional feature-based CNC toolpath module.

A Running the test suite

From the repository root:

```
# Pure-NumPy core (no Blender needed)
python3 reverse_mesh/tests/test_fitting.py
python3 reverse_mesh/tests/test_step.py
python3 reverse_mesh/tests/test_decompose.py
python3 reverse_mesh/tests/test_solidfit.py

# OCCT kernel round-trip (needs cadquery-ocp)
python3 reverse_mesh/tests/test_occ_export.py

# Blender integration (headless)
blender --background --python reverse_mesh/tests/blender_smoke.py
```

B License

BlenderMesh2STEP is free software under the GPL-3.0-or-later license.